

CHAYAN KUMAR

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Summary

Software Engineer with expertise in autonomous ML pipelines, data validation, and Agentic AI architectures. Demonstrated success in orchestrating LLMs for complex automated systems and engineering real-time hardware-software integrations. Strong foundation in AWS, deep learning, and building robust, data-driven solutions to solve real-world engineering challenges.

Experience

AI/ML Engineer **May 2026 – Present**
Lyptus Technologies *Bangalore, India*

- Engineered an end-to-end agentic workflow to autonomously generate and verify SystemVerilog/RTL code from natural language specs
- Architected a strict state-machine orchestration pipeline (FastAPI, Python) integrating LLMs (Z.ai, OpenAI) to generate test plans, code, and architecture documents
- Built an automated repair loop triggering on Verilator failures, analyzing error logs and dynamically patching source code without human intervention
- Developed a real-time (30Hz) AI mirroring system using MediaPipe and ROS2 to map human boxing poses from video onto dual humanoid robots
- Engineered a systematic 9-DOF joint calibration workflow, debugging complex URDF mirrored-axis kinematics to achieve structurally accurate combat animations in RViz2

Machine Learning Associate **October 2024 – March 2026**
Amazon — AWS Bedrock Team *Bangalore, India*

- Improved data quality for AI/ML workflows by validating and refining large-scale datasets, improving overall data consistency
- Built and maintained AWS QuickSight dashboards to track operational metrics and provide insights to stakeholders
- Developed Python automation scripts for data validation, cross-referencing, and workflow improvements in AWS SageMaker
- Collaborated with cross-functional teams to identify, debug, and resolve workflow issues, ensuring smooth data flow
- Shared best practices on data handling, prompt engineering, LLM observability, and LLM output evaluation with team members

Software Engineer Trainee **July 2024 – October 2024**
Revature *Pune, India*

- Trained in Java, J2EE, Spring Boot, and Maven for backend application development
- Built backend modules following MVC architecture and industry-standard coding practices
- Wrote SQL queries, managed JSON/YAML configurations, and practiced version control using Git
- Worked in an Agile environment, participating in sprint planning, code reviews, and CI/CD practices

Patent/Academic Project

Fruit Quality Detection and Classifier | *Python, CNN, TensorFlow, Keras*

- Built a CNN-based image classification model to detect fruit quality using labeled datasets
- Trained the model on labeled image datasets containing multiple fruit categories and quality classes
- Evaluated model performance using validation metrics and improved generalization through data handling techniques

Skills

Programming & Core CS: Python, C++, Data Structures & Algorithms

ML, AI & Data: TensorFlow, PyTorch, Keras, scikit-learn, RAG, GenAI, Agentic AI, MediaPipe, Pandas, NumPy

Cloud, Tools & Frameworks: AWS QuickSight, Git, Docker, Linux, ROS2, FastAPI

Education

Bachelor of Technology in Computer Science — CGPA: 7.8/10

Dr. APJ Abdul Kalam Technical University

2020 – 2024

Lucknow, Uttar Pradesh